Digital Hardware Design Ramp-up Project

January 2024

Changelog

Version 1.1

- Added this Changelog page to track changes.
- Bug fixes in Verilog code (*Listing 1*).
- Modified point 4 to enforce behavioral modeling in Verilog.

Introduction

It is Kaalrav season and SMIT is once again bustling with people from far and wide. The stage is set for one of the greatest cultural extravaganza in the region. There are lessons learned from the previous edition of Kaalrav and the institution's cultural secretary have laid down plans to mitigate those issues before the new edition kicks off.

Figure 1: SMIT campus *(Courtesy: Google Earth)*

It has been estimated that the number of attendees this year would at least be twice as large compared to the yesteryear. This data has triggered various crowd control activities and drills as part of the preparation for the big day. One of the major challenges faced during the previous year was managing high amount of vehicular and crowd traffic around the campus. To alleviate this problem, the organizers have proposed installation of traffic lights in a key junction in the campus. The system should ensure execution of smooth flow of pedestrian and vehicular traffic in the area. The organizing team after employing a few systems engineers have proposed a design to tackle the problem. Your team has been selected for the hardware implementation of the algorithm for smooth traffic flow.

Informal Description of Desired Traffic Light System

The road intersection and traffic light system are depicted schematically in Figure 2[1]. It is a T-junction where the major road is the East—West (OBH—MR) road. The road from the guest house (PMC) intersects this road. The normal cycle of light is the following[2]:

[1]The algorithm is open for changes that are inevitable for the design

[2]Note that the description given in this section is not the most optimal way to control traffic. That is the job of traffic engineers. We are electronics engineers and let's not worry about the algorithm for now ;)

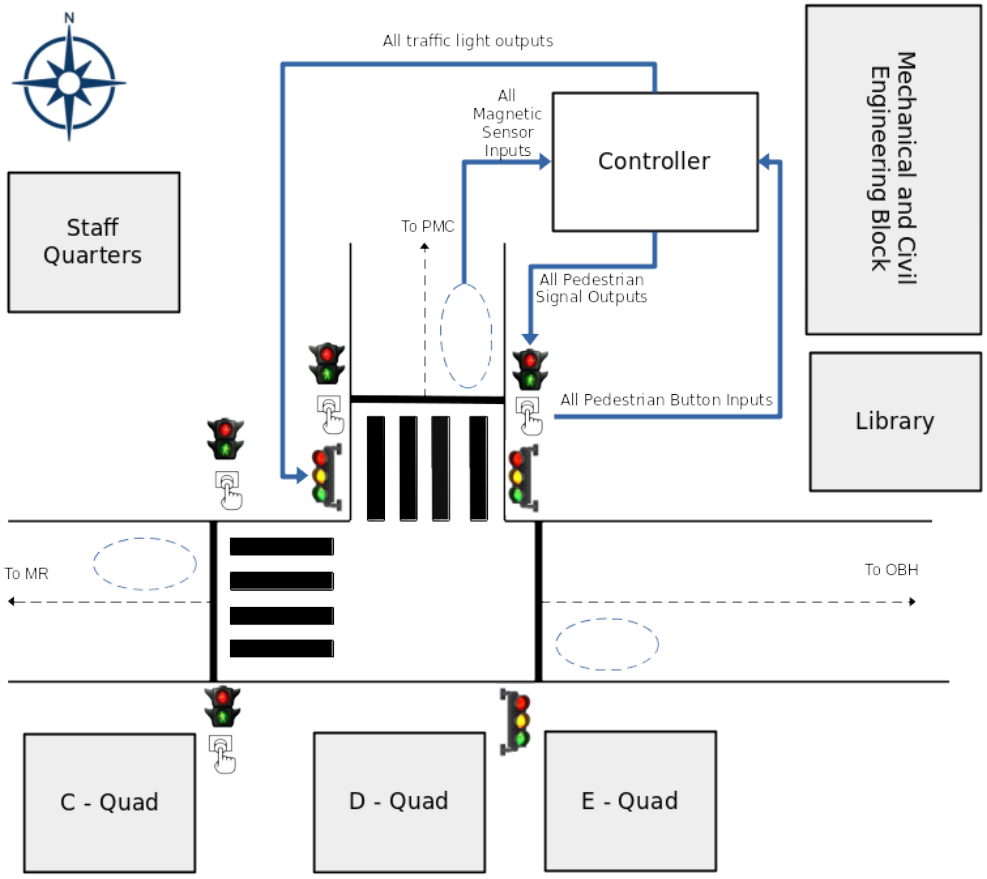

Figure 2: Layout of the intersection

- Allow East—West traffic (both directions): Green for 20 seconds, 4 seconds yellow and 2 seconds all red.
- Allow MR—PMC traffic: 10 seconds green, 4 seconds yellow and 2 seconds all red.
- Allow OBH—PMC traffic: 10 seconds green, 4 seconds yellow and 2 seconds all red.

The "all red" periods are safety periods where all signals turn red for 2 seconds before continuing to the next cycle. The green lights also indicate the direction to which traffic is allowed in a cycle. Only one of the three cycles above can be active at a time pertaining to vehicular motion.

The system also facilitates three types of 'intelligent' behaviour. Firstly, it can, on request add a pedestrian sequence to the normal sequence. There are points, as depicted in *Figure 2* where pedestrians can press a button that affects the sequence of the traffic lights. When pressed at any point in the sequence, after the "all-red" of the current sequence, the pedestrian light go green for 10 seconds and all other road

traffic lights will turn red. After this, the pedestrian lights glow yellow for 6 seconds, then a mandatory 2 seconds "all red" and then the cycle resumes. Once a pedestrian button is pressed, all further inputs to the buttons are ignored during the pedestrian sequence.

All three incoming roads have magnetic sensors which detects the presence of car in the incoming lane. The second intelligent behaviour is that if there are no vehicles waiting on an incoming road, then its corresponding green/yellow/all-red sequence is skipped. This also means that if there are not cars on all incoming roads, all lights will be red i.e., the pedestrian traffic is given preference.

Finally, the third intelligent behavior is that if a particular traffic flow is given green because there was at least one car waiting, the light will transition to yellow and then all-red when the magnetic sensor no longer senses any traffic.

Design Notes

Your goal is to build the box labeled 'Controller' in *Figure 2*. It takes Boolean inputs from all magnetic sensors and pedestrian buttons. We assume that all pedestrian buttons are OR'ed together. The 'Controller' produces outputs that controls the lights on the pedestrian and vehicular traffic lamps with directional information.

The design has to be implemented on hardware. Your group is tasked with design and verification of the model in Verilog. A Verilog model of the clock and timer circuit to be used in the design is attached at the end of this document *(See Listing 1)*.

The design should absorb all the characteristics described in the previous section including trivial ones. For example, when traffic signal is green, it is automatically inferred that the pedestrian light should be red, even though not explicitly specified.

Tasks and instructions for the Working Group

The working group should follow the following task list to ensure continuous and comprehensive evaluation of the work:

1. This assignment should be a group effort. *Ideate, Communicate, Contribute!*
2. All artefacts related to this project should be pushed to a GitHub repo. This will help everyone to evaluate progress.
3. The first checkpoint to this assignment is crafting the design document. The document should feature architectural and micro-architectural details of the design. It should also employ the use of state diagram(s) to show how different events affect the states of the architecture. The design document will be evaluated by systems engineers. Once the document is perfected, the team can go ahead with implementation of the design. *Divide and Conquer!*

4. The design should be implemented using Verilog HDL. Only use behavioral modeling. Ensure functionality first before you think of over-optimizing. The design should be simulated using Verilator. Usage of Xilinx Vivado or any other FPGA CAD tools is discouraged. Use your favourite text editor like Vim, Notepad++, VS Code etc. Integration of Verilator to the work flow can be automated using Bash scripting or Makefiles.

5. Once the design has matured enough, the group shall start writing a test plan. The test plan is an additional document which features a checklist of all reasonable corner cases and inputs used for the sake of verifying the design. The document shall be validated before the testbench is implemented in Verilog.

6. By the end of this assignment, the team should have designed and simulated the design with test inputs that captures the maximum number of corner cases. The VCD file generated would be an added submission article to convince the systems engineers that the design can be forwarded for synthesis related activities.

Additional Material

Listing 1: Clock and Timer Verilog Code

```
/**
 * This module generates the clock and seconds timer.
 * The clock is assumed to tick every 1 ms.
**/
module clock_timer(
                    output wire        clk,
                    output wire [32:0] time32
                  );

// internal registers that store state
reg r_clk;
reg [32:0] r_time32;

// initialize registers
initial begin
        r_clk = 1'b0;
        r_time32 = 32'b0;
end

// assume clk toggles every 1ms for #1.
// hence second timer increments by 1 after every #1000
always #1 r_clk = ~r_clk;
always #1000 r_time32 = r_time32 + 1;

// wire out the registers
assign clk = r_clk;
assign time32 = r_time32;

endmodule
```